

THE EXPONENT SPECTRUM OF PERFECT-POWER VALUES OF POLYNOMIALS, WITH COMPLETE RUNGE ENUMERATION AND A MACHINE-CHECKED KERNEL

PERFECTPOWER PROJECT

ABSTRACT. For $F \in \mathbb{Z}[x]$ and $d \geq 2$ let $A(N)$ be the number of $1 \leq n \leq N$ for which $F(n)$ is a perfect d -th power. We make explicit, with constants and a decision procedure, the classification of (F, d) into four types that follows from Siegel's theorem on integral points, combined with classical Pell and valuation counting. The reduction to Siegel is an Euler-characteristic computation, $\chi = d'(1 - S)$, for the curve $y^d = F(x)$; it recovers the exceptional patterns of LeVeque (1964). The types are decidable from the root multiplicities of F . In the four types, $A(N) = N$; $A(N) = \kappa N^{1/t} + O(1)$ with $t \mid d$; $A(N) = \kappa \log N + O(1)$; or $A(N) = O(1)$. The constants κ are explicit. The first three types are handled by elementary arguments. The last rests on Siegel's theorem, so it is ineffective. We regard the classification as a synthesis rather than a new theorem. Consequently the growth exponent $\limsup \log(1 + A(N))/\log N$ lies in $\{0, 1\} \cup \{1/t : t \mid d, t > 1\}$, and $A(N) = O(N^{1/p})$ for p the least prime factor of d whenever F is not a d -th power in $\mathbb{Z}[x]$. In the Runge case ($d \mid \deg F$ and $\text{lc } F$ a d -th power) we give a complete enumeration of the hit set by exact root isolation, together with a pipeline that emits Lean proofs of complete hit sets; these are the parts we consider possibly new. We describe the dependence on a shift $F = S + k$, and give Dirichlet-series and heat-kernel asymptotics for every type. Several results are verified in Lean 4 with Mathlib. They include the rigid dichotomy and the rigid 0–1 law, the counting core of the radical type, the finiteness of Pell orbit representatives, the combinatorial half of the Euler-characteristic computation, and a function-field Pillai inequality. The generated certificates are checked by a verified reflective checker.

1. INTRODUCTION

Let $F \in \mathbb{Z}[x]$ and $d \geq 2$. Call $n \geq 1$ a *hit* if $F(n) = m^d$ for some $m \in \mathbb{Z}$, and let $A(N) = \#\{1 \leq n \leq N : n \text{ is a hit}\}$. The density $\lim A(N)/N$ always exists: it is 1 if $F = G^d$ with $G \in \mathbb{Z}[x]$ and 0 otherwise. This follows, for instance, from Boshernitzan's equidistribution theorem for Hardy fields together with a separate treatment of the “rigid” case, and this was the starting point of the project.

Density zero is a weak statement. The hit sets of $4n + 1$, of $2n^2 + 1$ and of $n^3 + 1$ (with $d = 2$) have respectively about $\sqrt{N}$ elements, about $0.567 \log N$ elements, and finitely many elements. The purpose of this paper is to make explicit that these three behaviours, together with density one, are the only possibilities. We show this is a consequence of Siegel's theorem [2] (through Theorem 4.4, which recovers LeVeque's patterns [7]) and classical counting, compute everything explicitly, and certify as much as possible by machine. We claim no priority for the classification itself.

Setup. Write $F = c \prod_i (x - \alpha_i)^{r_i}$ over $\overline{\mathbb{Q}}$ with distinct α_i and $c = \text{lc } F$. Put $t_i = d/\gcd(d, r_i)$. The multiset $\{t_i\}$ is the t -*profile* of (F, d) . It is determined by Yun's squarefree decomposition $F = c \prod_j S_j^{j_j}$: every root of S_j has $t = d/\gcd(d, j)$. No factorisation is needed.

Theorem 1.1 (Atlas). *Let $F \in \mathbb{Z}[x]$ be nonconstant and $d \geq 2$. Exactly one of the following holds.*

- (i) **Power type:** *all $t_i = 1$. Then $A(N) = N$ if c is an integer d -th power, and otherwise the hits are exactly the positive integer roots of F .*
- (ii) **Radical type:** *the t -profile is $\{t, 1, \dots, 1\}$ with $t \geq 2$. Then $A(N) = \kappa N^{1/t} + O(1)$ with $\kappa \geq 0$ given by (1).*
- (iii) **Pell type:** *the t -profile is $\{2, 2, 1, \dots, 1\}$. Then $A(N) = \kappa \log N + O(1)$ with $\kappa \geq 0$ given by (2).*
- (iv) **Finite type:** *all remaining cases. Then the hit set is finite.*

Corollary 1.2 (Exponent spectrum). *Let $\alpha(F, d) = \limsup_N \log(1 + A(N))/\log N$. Then $\alpha(F, d) \in \{0, 1\} \cup \{1/t : t \mid d, t \geq 2\}$, and each value occurs. If F is not the d -th power of a polynomial in $\mathbb{Z}[x]$, then $A(N) = O(N^{1/p})$, where p is the least prime factor of d . In particular $A(N) = O(\sqrt{N})$ always, and $A(N) = O(N^{1/3})$ for odd d . These bounds are attained.*

Proof. In type (ii) the integer $t = d/\gcd(d, r)$ divides d and exceeds 1, so $t \geq p$. Types (iii) and (iv) have exponent 0. The monomial x^r with $\gcd(r, d) = d/t$ has $A(N) = \lfloor N^{1/t} \rfloor$ by Theorem 3.1. $\square$

Parts (i)–(iii) of Theorem 1.1 are proved in §§2–4. Part (iv) is Theorem 4.4, which derives it from Siegel’s theorem in its standard form. LeVeque’s theorem [7] states the same exceptional patterns. We had it only from secondary sources, so we do not use it. We therefore obtain a second proof of the density 0–1 law, independent of equidistribution. It depends on Siegel’s theorem through Theorem 4.4, whose geometric step is a paper proof that has not been refereed. It is quantitative in the rate but *ineffective* in the finite type, so it cannot replace the equidistribution argument where effectivity matters. Section 5 makes type (iv) effective in the Runge case. Section 6 treats shifts, §7 treats transforms, and §10 describes the Lean formalisation and the computations.

2. THE POWER TYPE

Theorem 2.1. *Suppose all root multiplicities of F are divisible by d , so that $F = cG^d$ with $G \in \mathbb{Q}[x]$ monic. If $c = b^d$ with $b \in \mathbb{Z}$, then $bG \in \mathbb{Z}[x]$ and every n is a hit. Otherwise $n \geq 1$ is a hit if and only if $F(n) = 0$.*

Proof. If $c = b^d$, then $(bG)^d = F \in \mathbb{Z}[x]$. Since $\mathbb{Z}[x]$ is integrally closed in $\mathbb{Q}[x]$ (Gauss’s lemma), $bG \in \mathbb{Z}[x]$. Now suppose c is not an integer d -th power. If $G(n) \neq 0$ and $F(n) = m^d$, then $c = (m/G(n))^d$. A rational number whose d -th power is an integer is an integer, so c would be an integer d -th power, a contradiction. Finally, $F(n) = 0 = 0^d$ whenever $G(n) = 0$. $\square$

Remark 2.2 (Grunwald–Wang). Take $F = 16x^8$ and $d = 8$. Since 16 is an 8-th power modulo every odd prime, $F(n)$ is an 8-th power residue modulo every prime for every n . Yet F has no hits. No sieve that uses residues modulo primes alone can prove density zero in general. Here the obstruction is modulo 32.

3. THE RADICAL TYPE

Galois-conjugate roots of F have equal multiplicities. A unique root with $t \geq 2$ is therefore rational, say $\alpha = u/v$ with $v \geq 1$ and $\gcd(u, v) = 1$. We can write

$$F = c(x - \alpha)^r G(x)^d, \quad G \in \mathbb{Q}[x] \text{ monic}, \quad t = d/\gcd(r, d) \geq 2.$$

Let $K = \lceil r/d \rceil$ and $c_1 = cv^{dK-r} \in \mathbb{Z}$. Let $g = \gcd(r, d)$ and $r' = r/g$, so $\gcd(r', t) = 1$. If $g \mid v_p(c_1)$ for all primes p , let $\tau_p \in [0, t)$ be defined by $\tau_p \equiv -(v_p(c_1)/g)r'^{-1} \pmod{t}$, and put $z_0 = \prod_p p^{\tau_p}$. A sign $\sigma \in \{\pm 1\}$ is *admissible* if d is odd, or if d is even and $c_1\sigma^r > 0$.

Theorem 3.1. *In the radical type, $n \geq 1$ is a hit if and only if either $F(n) = 0$, or $g \mid v_p(c_1)$ for every p and $vn - u = \sigma z_0 w^t$ for an admissible sign σ and an integer $w \geq 1$. Let $\varrho = \#\{w \bmod v : z_0 w^t + u \equiv 0 \pmod{v}\}$. If $+1$ is admissible and the divisibility condition holds, then*

$$(1) \quad A(N) = \kappa N^{1/t} + O(1), \quad \kappa = \frac{\varrho}{v} \left(\frac{v}{z_0} \right)^{1/t}.$$

Otherwise $A(N) = O(1)$.

Proof. Let $F(n) \neq 0$. We claim that n is a hit if and only if $c(n - \alpha)^r \in \mathbb{Q}^{\times d}$. If $F(n) = m^d$, divide by $G(n)^d$. Conversely, if $c(n - \alpha)^r = \xi^d$, then $F(n) = (\xi G(n))^d$ is an integer and a rational d -th power, hence an integer d -th power.

With $z = vn - u \neq 0$, multiplying by the d -th power v^{dK} shows that the condition is equivalent to $c_1 z^r$ being the d -th power of an integer. An integer $X \neq 0$ is a d -th power if and only if $d \mid v_p(X)$ for every p , together with $X > 0$ when d is even. For $X = c_1 z^r$ the valuation condition is $v_p(c_1) + rv_p(z) \equiv 0 \pmod{d}$. This is solvable exactly when $g \mid v_p(c_1)$, and then it is equivalent to $v_p(z) \equiv \tau_p \pmod{t}$. The positive integers with these valuations are exactly $z_0 w^t$ with $w \geq 1$.

For the count, the sign $\sigma = +1$ contributes $n = (z_0 w^t + u)/v$ with w in one of ϱ residue classes modulo v and $z_0 w^t \leq vN - u$. That gives $\varrho v^{-1}((vN - u)/z_0)^{1/t} + O(1)$ hits. The sign $\sigma = -1$ forces $n < \alpha$, and there are finitely many zeros of F . $\square$

Example 3.2. • $4n + 1$ with $d = 2$: $\kappa = 1$.

- $2n - 5$ with $d = 2$: $\kappa = 2^{-1/2}$.
- $3n + 5$ with $d = 3$: $\kappa \approx 0.4807$.
- $12n^2$ with $d = 3$: the hits are $n = 12w^3$, so $\kappa = 12^{-1/3}$.
- $3n + 2$ with $d = 6$ and $16n^3$ with $d = 6$: no hits.

4. THE PELL TYPE

Here $d = 2e$, and the two special roots have multiplicities that are odd multiples of e . They are either both rational or conjugate. Their monic product $W \in \mathbb{Q}[x]$ has distinct roots, and $F = cW^e H^d$ with $H \in \mathbb{Q}[x]$ monic.

Theorem 4.1. *Let $\Gamma = \{\gamma \in \mathbb{Q} : \gamma^e = c\}$, so that $|\Gamma| \leq 2$. For $\gamma = \gamma_1/\gamma_2 \in \Gamma$ in lowest terms, let L be the common denominator of W and put $P_\gamma = \gamma_1 \gamma_2 L \cdot LW \in \mathbb{Z}[x]$. Then $n \geq 1$ is a hit if and only if $F(n) = 0$ or $P_\gamma(n)$ is a perfect square for some $\gamma \in \Gamma$.*

Proof. For $F(n) \neq 0$, n is a hit if and only if $cW(n)^e = y^{2e}$ with $y \in \mathbb{Q}^\times$. In that case $\gamma = y^2/W(n)$ satisfies $\gamma^e = c$. Conversely, if $\gamma W(n) = y^2$ with $\gamma^e = c$, then $cW(n)^e = y^{2e}$. Multiplying by $(\gamma_2 L)^2$ turns “ $\gamma W(n)$ is a rational square” into “ $P_\gamma(n)$ is a perfect square”. $\square$

Lemma 4.2. *Let $P = An^2 + Bn + C \in \mathbb{Z}[x]$ with $A \neq 0$ and $\Delta = B^2 - 4AC \neq 0$.*

- (a) *If $A < 0$ or A is a square, then $P(n)$ is a square for only finitely many n .*
- (b) *If $A > 0$ is not a square, let $\varepsilon = x_1 + y_1 \sqrt{4A}$ be the fundamental solution of $x^2 - 4Ay^2 = 1$. For each ε -orbit $\mathcal{O}$ of solutions $\eta = X + Y\sqrt{4A} > 0$ of $X^2 - 4AY^2 = \Delta$, let $\pi_{\mathcal{O}}$ be the period of $(X, Y) \bmod 2A$ along the orbit, and let $g_{\mathcal{O}}$ be the number of steps in one period with $X \equiv B \pmod{2A}$. Then*

$$(2) \quad \#\{n \leq N : P(n) = \square\} = \kappa \log N + O(1), \quad \kappa = \frac{1}{\log \varepsilon} \sum_{\mathcal{O}} \frac{g_{\mathcal{O}}}{\pi_{\mathcal{O}}}.$$

Proof. We have $4AP(n) = X^2 - \Delta$ with $X = 2An + B$. So $P(n) = m^2$ if and only if $X^2 - 4Am^2 = \Delta$, and $n \mapsto X$ is a bijection onto the class $B \pmod{2A}$.

In (a), $A < 0$ bounds X . If $A = s^2$, the factorisation $(X - 2sm)(X + 2sm) = \Delta$ leaves at most $2\tau(|\Delta|)$ possibilities.

In (b) there are finitely many orbits (Nagell; in Lean, `pell_orbits_exhaust`: descent by ε^{-1} lands every solution with $X > 0$, $Y \geq 0$ in the box $4AY_0^2 \leq |\Delta|x_1^2$). A large hit corresponds to exactly one large $\eta = \eta_0\varepsilon^k$ in a positive orbit with $X \equiv B \pmod{2A}$. The step $(X, Y) \mapsto (x_1X + 4Ay_1Y, y_1X + x_1Y)$ has determinant 1, so it is invertible modulo $2A$ and the residues are purely periodic. Finally $\log X_k = k \log \varepsilon + O(1)$. $\square$

Example 4.3. For $2n^2 + 1$ we get $\kappa = 1/\log(3 + 2\sqrt{2}) = 0.56729\dots$. The exact counts at $N = 10^6, 10^{12}, 10^{24}, 10^{48}$ are 8, 16, 31, 63. The square triangular numbers $n(n+1)/2$, reduced to $4n^2 + 4n$ by Remark 4.6, have the same κ . For the rigid polynomials $n^2 + n$ and $n^2 + 1$, A is a square and there are no hits.

Theorem 4.4 (Geometry of $y^d = F(x)$). *Let $F = c \prod_{i=1}^s (x - \alpha_i)^{r_i}$ with distinct $\alpha_i \in \overline{\mathbb{Q}}$. Put $g = \gcd(d, r_1, \dots, r_s)$, $d' = d/g$, $t_i = d/\gcd(d, r_i)$ and $S = \sum_i (1 - 1/t_i)$. Over $\overline{\mathbb{Q}}$, the curve $y^d = F(x)$ splits into g irreducible components. Each is a cyclic cover of $\mathbb{P}^1$ of degree d' with $n_\infty = \gcd(d', \deg F/g)$ points over infinity, and the affine part of each component has Euler characteristic*

$$\chi = 2 - 2g_C - n_\infty = d'(1 - S).$$

Consequently $y^d = F(x)$ has only finitely many integral points when $S > 1$. Moreover $S \leq 1$ holds exactly for the power, radical and Pell profiles.

Proof. Capelli's theorem gives the irreducibility of each factor $Y^{d'} - \zeta c^{1/g} \prod (x - \alpha_i)^{r_i/g}$. By Kummer theory, at a place where $y^{d'}$ has valuation v there are $\gcd(d', v)$ points, each with ramification index $d'/\gcd(d', v)$. This gives $\gcd(d', r_i/g) = d'/t_i$ points over α_i , $\gcd(d', \deg F/g)$ points over ∞ , and d' unramified points over every other x . Riemann–Hurwitz, $2g_C - 2 = -2d' + \sum_i (d' - d'/t_i) + (d' - n_\infty)$, then yields the formula for χ . If $S > 1$ then $\chi < 0$, which is Siegel's condition $2g_C - 2 + n_\infty > 0$ (Hindry–Silverman, Theorem D.9.1).

Siegel's theorem applies to the normalisation of each component. Integral points avoid the finitely many singular points. On the normalisation they have coordinates with denominators bounded by a fixed integer, because the coordinate ring is the integral closure of $K[x]$. So they are S -integral for a suitable finite S .

Finally, each special root contributes at least $\frac{1}{2}$ to S . Hence $S \leq 1$ allows at most one special root with arbitrary t , or two with $t = 2$. $\square$

Remark 4.5 (Checks of Theorem 4.4). The last paragraph of the proof is compiled in Lean (`S.le_one_iff`). Lean also checks by kernel evaluation (`profile_table_ok`) that for every $2 \leq d \leq 12$ and every multiplicity profile of degree at most 12, the genus from Riemann–Hurwitz is a nonnegative integer and $\chi < 0$ exactly outside the three exceptional profiles. Independently, Singular's normalisation computes g_C over the same range, Sage's function fields compute n_∞ where feasible, and we compare both with the formula. The Kummer and Riemann–Hurwitz step itself remains a paper argument.

Proof of Theorem 1.1. The t -profiles in (i)–(iv) are mutually exclusive and exhaust all possibilities. Parts (i)–(iii) are Theorems 2.1, 3.1 and 4.1 together with Lemma 4.2. Part (iv) is Theorem 4.4, which recovers LeVeque's exceptional patterns [7] from Siegel's theorem. $\square$

Remark 4.6 (Integer-valued polynomials). If $F \in \mathbb{Q}[x]$ is integer-valued with coefficient denominator L , then F and $L^d F \in \mathbb{Z}[x]$ have the same hits. This brings binomial coefficients into scope. For example, $\binom{n}{3} = m^2$ reduces to integral points on $Y^2 = X^3 - 36X$ via $X = 6(n-1)$, $Y = 36m$,

and $\binom{n}{2} = m^3$ reduces to $Y^2 = X^3 + 1$ via $X = 2m$, $Y = 2n - 1$. Lean proves both reductions. Given the integral points of the two curves, computed by Sage (rank proved, saturated basis, elliptic-logarithm sieving) and taken as explicit hypotheses, the hit sets are $\{1, 2, 3, 4, 50\}$ and $\{1, 2\}$.

Example 4.7 (Sums of powers). Let $S_k(n) = 1^k + \cdots + n^k$. Classifying $(L^d S_k, d)$ for $k \leq 10$ and $d \leq 6$ recovers Schäffer's theorem [11]: there are infinitely many solutions exactly for $(k, d) = (1, 2), (3, 2), (3, 4), (5, 2)$. The first is of Pell type with $\kappa = 1/\log(3 + 2\sqrt{2})$, and the second is of power type (Nicomachus). The third is of Pell type with the same hits as $(1, 2)$. The fourth is of Pell type, with hits $1, 13, 133, 1321, \dots$ and $\kappa = 1/\log(5 + 2\sqrt{6})$. Among the finite cases, a scan finds only Lucas's $n = 24$ for $(2, 2)$.

5. COMPLETE ENUMERATION IN THE RUNGE CASE

Suppose $\deg F = dq$ and $\text{lc } F = b^d$ with $b \in \mathbb{Z}$ (the *rigid* case), and that F is not a d -th power. Matching the top $q + 1$ coefficients gives a truncated root $Q = P/D$ with $P \in \mathbb{Z}[x]$, $D \geq 1$ and $\text{lc } Q = b$, such that $R = F - Q^d \neq 0$ has degree $r < (d - 1)q$. Qualitatively this shows that the hit set is finite. The Lean theorem `rigid_dichotomy` checks this.

Theorem 5.1. *For $x_0 \geq 1$ put*

$$a(x_0) = |b| - \sum_{i < q} |Q_i| x_0^{i-q}, \quad C(x_0) = \sum_{i \leq r} |R_i| x_0^{i-r}, \quad T(x_0) = \left\lfloor \frac{D C(x_0) x_0^{r-(d-1)q}}{a(x_0)^{d-1}} \right\rfloor.$$

Suppose $a(x_0) > 0$ and, when d is odd, $C(x_0)x_0^r < a(x_0)^{dq}$. Then every hit $n \geq x_0$ is a root of one of the nonzero integer polynomials

$$G_t = D^d F - (P + t)^d, \quad |t| \leq T(x_0).$$

In particular the number of hits is at most $x_0 - 1 + (2T(x_0) + 1)(d - 1)q$.

Proof. For $n \geq x_0$ we have $|Q(n)| \geq a(x_0)n^q > 0$ and $|R(n)| \leq C(x_0)n^r$. Let $F(n) = m^d$ and pick $y = \pm m$ with $y^d = F(n)$ and $yQ(n) \geq 0$. For odd d the sign condition forces $y = m$ to work, because $|R(n)| < |Q(n)|^d$ gives $F(n)$ the sign of $Q(n)$. Write $y = sQ(n)$ with $s \geq 0$. Then $|s - 1| \leq |s^d - 1| = |R(n)|/|Q(n)|^d$, so

$$|Dy - P(n)| \leq D|R(n)|/|Q(n)|^{d-1} \leq D C(x_0) x_0^{r-(d-1)q} / a(x_0)^{d-1}.$$

Hence $t = Dy - P(n) \in \mathbb{Z}$ satisfies $|t| \leq T(x_0)$ and $G_t(n) = 0$. Finally, $G_0 = D^d R \neq 0$, and for $t \neq 0$ the polynomial G_t has degree $(d - 1)q$ with leading coefficient $-dt(Db)^{d-1}$. $\square$

Since $T(x_0) \rightarrow 0$, the following algorithm is complete. Scan up to the first admissible x_0 . Then walk dyadic blocks $[x_0, 2x_0)$, and in each block either scan it or isolate the integer roots of all G_t with $|t| \leq T(x_0)$ by Sturm sequences, whichever is cheaper. Stop when $T(x_0) = 0$, where only G_0 remains. If F is rigid for some divisor $d' \geq 2$ of d , the same procedure applies, because d -th powers are d' -th powers. By Capelli's theorem this covers the leading-term form of Runge's criterion, up to a trivially negative case.

Example 5.2. $1 + n + n^2 + n^3 + n^4$ is a square for $n \geq 1$ only at $n = 3$ (Ljunggren). For the following pairs, the product of k consecutive positive integers is never a d -th power; each is a fixed instance of the Erdős–Selfridge theorem, and each takes at most 0.03 s:

k	d
4	2, 4
6	2, 3, 6
8	2, 4, 8
10	2, 5
12	2, 3, 4, 6

The earlier cutoff-only method for these pairs needed cutoffs as large as $3.9 \cdot 10^{10}$.

6. SHIFTS

Corollary 6.1. *Let $S \in \mathbb{Z}[x]$ with $\deg S = M \geq 1$, and let $\mathcal{K}(S)$ be the set of integer critical values $-S(\xi)$, where $S'(\xi) = 0$. It has at most $M - 1$ elements, and it is the set of integer roots of $\text{Res}_x(S + k, S')$. For $k \notin \mathcal{K}(S)$, the polynomial $S + k$ is squarefree and is:*

- of radical type if $M = 1$;
- of Pell type if $M = 2$ and $d = 2$;
- of finite type otherwise.

In particular, if $M \geq 3$ then $S(n) + k$ has finitely many hits for all but at most $M - 1$ values of k , and at most one k gives density one.

For example, $n^3 + k$ with $d = 2$ has finitely many hits for every $k \neq 0$; these are the Mordell curves. For $n^3 - 3n + k$ with $d = 2$ the exceptional shifts are $k = \pm 2$, and $n^3 - 3n + 2 = (n - 1)^2(n + 2)$ has $\kappa = 1$.

7. TRANSFORMS

Let X be the hit indicator, $Z_X(s) = \sum X(n)n^{-s}$ and $K_X(\tau) = \sum X(n)e^{-\tau n}$.

Theorem 7.1. *If $A(x) = \kappa x^\beta + O(1)$ with $0 < \beta \leq 1$, then $Z_X(s) - \kappa s/(s - \beta)$ is holomorphic on $\Re s > 0$, and $K_X(\tau) = \kappa \Gamma(1 + \beta) \tau^{-\beta} + O(1)$. If $A(x) = \kappa \log x + O(1)$, then $Z_X(s) - \kappa/s$ is holomorphic on $\Re s > 0$, and $K_X(\tau) = \kappa \log(1/\tau) + O(1)$.*

Proof. Use Abel summation, $Z_X(s) = s \int_1^\infty A(x)x^{-s-1} dx$ and $K_X(\tau) = \tau \int_0^\infty A(x)e^{-\tau x} dx$. The main terms integrate explicitly, and a bounded error contributes a function that is holomorphic on $\Re s > 0$, respectively bounded. $\square$

By Theorem 3.1 and Lemma 4.2 these hypotheses hold exactly in the radical and Pell types. When $\alpha = 0$ in the radical type, one gets $Z_X(s) = z_0^{-s} \zeta(ts)$ plus a Dirichlet polynomial.

In the Pell type the second term of K_X oscillates. Put $\beta = -B/(2A)$. Apart from finitely many hits, the hits split into classes $n_j = \alpha E^j + \beta + O(E^{-j})$, one for each positive orbit and good residue class, with $E = \varepsilon^{\pi \circ}$. The shift β comes from $X = 2An + B$.

Theorem 7.2. *In the Pell type,*

$$K_X(\tau) = e^{-\tau \beta} \sum_{\text{classes}} \left(\frac{\log(1/(\tau \alpha)) - \gamma}{\log E} + \frac{1}{2} + \Phi_E(\log \tau \alpha) \right) + c_{\text{int}} + O(\tau),$$

where $\Phi_E(u) = \frac{2}{\log E} \Re \sum_{m \geq 1} \Gamma(2\pi i m / \log E) e^{-2\pi i m u / \log E}$ is $\log E$ -periodic and $c_{\text{int}} \in \mathbb{Z}$ counts the finitely many hits outside the classes minus the model terms missing from the hit set. Hence $K_X(\tau) = \kappa \log(1/\tau) + C_0 + \sum \Phi_E(\log \tau \alpha) + O(\tau \log(1/\tau))$. The remainder is $O(\tau)$ when $B = 0$ and not in general.

Proof. The Mellin transform of $\sum_j e^{-\lambda E^j}$ is $\Gamma(s)/(1-E^{-s})$. The double pole at 0 gives the main term, the poles at $2\pi im/\log E$ give Φ_E , and shifting to $\Re s = -\frac{1}{2}$ leaves $O(\lambda)$. Writing $e^{-\tau n_j} = e^{-\tau\beta} e^{-\tau\alpha E^j} (1 + O(\tau E^{-j}))$ costs $O(\tau)$ in total. For the two-term form, $(e^{-\tau\beta} - 1)\kappa \log(1/\tau) = -\beta\kappa \tau \log(1/\tau) + O(\tau^2 \log(1/\tau))$. $\square$

An earlier version stated the unshifted form with remainder $O(\tau)$, which is false for $B \neq 0$. Numerically, for $2n^2 + 2n$ the unshifted residual divided by τ grows from 2.8 to 8.1 as τ decreases from 10^{-4} to 10^{-12} . The shifted residual divided by τ stays at 0.25.

8. EXPONENTIAL SEQUENCES

Theorem 8.1. *Let $a \geq 2$, $c \neq 0$, $d \geq 2$, and write $a = b^s$ with s maximal. The integers $n \geq 1$ for which ca^n is a d -th power form either the empty set or one residue class modulo $L = d/\gcd(d, s)$. Hence the density is 0 or $\gcd(d, s)/d$.*

Proof. For d even and $c < 0$ there are no hits. Otherwise the condition is $v_p(c) + n v_p(a) \equiv 0 \pmod{d}$ for all primes p . Each such congruence is either a solvability condition (when $p \nmid a$) or it fixes n modulo $d/\gcd(d, v_p(a))$. The least common multiple of these moduli is $d/\gcd(d, s)$. $\square$

Thus the polynomial and exponential cases have dual discrete spectra: exponents $1/t$ for polynomials and densities $1/L$ for exponentials, with $t, L \mid d$. For a nonzero shift k , $ca^n + k = m^d$ has finitely many solutions, by Thue's theorem for $d \geq 3$ and by the S -unit theorem for $d = 2$. An example is Ramanujan–Nagell, $2^n - 7 = m^2$ only for $n = 3, 4, 5, 7, 15$.

9. FUNCTION FIELDS AND THE ABC-CONDITIONAL LAYER

Hall's and Pillai's conjectures are open, and nothing here proves them. Over $k[x]$ with $\text{char } k = 0$ their analogues follow from the Mason–Stothers theorem, which Mathlib provides.

Theorem 9.1 (Function-field Pillai). *Let $f, g \in k[x]$ be coprime, with f nonconstant, $a, b \geq 1$ and $f^a \neq g^b$. Then $a \deg f + 1 \leq \deg f + \deg g + \deg(f^a - g^b)$ and $b \deg g + 1 \leq \deg f + \deg g + \deg(f^a - g^b)$. If $a, b \geq 2$ and $a \deg f = b \deg g = n$, then $\deg(f^a - g^b) \geq n(1 - 1/a - 1/b) + 1$. For $(a, b) = (3, 2)$ this is Davenport's bound $2 \deg(f^3 - g^2) \geq \deg f + 2$, which is attained by $f = x^2 + 2$, $g = x^3 + 3x$.*

Proof. Apply Mason–Stothers to $f^a - g^b - (f^a - g^b) = 0$. The radical of $f^a g^b (f^a - g^b)$ has degree at most $\deg f + \deg g + \deg(f^a - g^b)$. The degenerate case, where all derivatives vanish, forces $f' = 0$. The balanced form follows by multiplying by ab . $\square$

Over $\mathbb{Z}$ we state abc as an explicit hypothesis $\text{ABC}(\varepsilon, C)$, never as an axiom, and derive Hall's inequality for coprime x, y and Pillai finiteness uniform in the exponents. All of this is in Lean (`pillai_polynomial`, `davenport`, `hall_of_abc`, `pillai_finite_of_abc`). A census of all integral points on $y^2 = x^3 + k$, $0 < |k| \leq 10^4$ (Sage), and of all pairs of perfect powers up to 10^{18} at distance at most 1000 records the numerical frontier. It is evidence, not proof.

10. FORMAL VERIFICATION AND COMPUTATION

The Lean 4 library in the repository compiles against Mathlib (tag `v4.20.0`). An audit script checks that 112 declarations depend only on `propext`, `Classical.choice` and `Quot.sound`; the current count is regenerated in the repository README. Among them:

- `rigid_dichotomy`: in the rigid case, either $F = G^d$ over $\mathbb{Z}$ or the hit set is finite.
- `rigid_zero_one`: in the rigid case the density exists and is 0 or 1.
- `twisted_finite`: if $D^d F = cH^d$ with c not a d -th power, every hit is a root of H .

- `monomial_count`, the valuation core of Theorem 3.1 (`mul_pow_isPow_iff_param`: the solutions are $z = z_0 w^t$), and its count `radical_count_bound`: $|vA(N) - RW| \leq 2Rv$ with R good residues modulo v .
- The Pell interfaces of Lemma 4.2: `quadratic_isHit_iff_norm`, `unitOrbit_periodic` (pure periodicity modulo any M), `goodClass_hits`, `pell_orbits_exhaust` (finitely many orbit representatives), geometric counting, and `pell_count_log`: $A(N) = O(\log N)$ for every quadratic. The exact constant κ is not yet compiled.
- `S_le_one_iff` and `profile_table_ok`: the combinatorial half of Theorem 4.4.
- `pell_hitSet_infinite` and `pell_hasDensity_zero` for $2n^2 + 1$; `runge_finite`; `power_type_finite`; `exp_hasDensity`.
- The reductions of $\binom{n}{2} = m^3$ and $\binom{n}{3} = m^2$ to elliptic curves, with the curves' integral points as hypotheses.
- 19 machine-generated theorems of the form “the hit set of F is exactly H ”. Seventeen use a Runge threshold with Taylor-shifted polynomials, checked by a verified reflective checker (`rungeCheck_sound`). Two, products of 10 consecutive integers ($d = 2$) and of 12 ($d = 4$), use *interval sandwiches*: $[1, \infty)$ is covered by intervals on which $(P + t)^d < D^d F < (P + t + 1)^d$. These two are now *reflective*. The certificate is a data literal, and a Boolean checker, whose soundness (`check_sound`) is proved once, recomputes every Taylor shift and power by kernel evaluation. This took the generated file from 768 KB to 12 KB. Replacing the interval test $\text{POS}(0) > \text{NEG}(w)$ by a Bernstein bound would save only 4 of 283 pieces, because single points at small n dominate.

The exact Python implementation (standard library only) classifies (F, d) , computes κ , enumerates hits structurally or by Theorem 5.1, and cross-checks every structural count against the defining scan. A modular sieve makes exact scans to 10^8 take seconds. Differential fuzzers compare the classifier with brute force under fixed seeds; they were contributed by an external reviewer. Every receipt is reproducible bit-for-bit, and `make verify` regenerates them and fails on any difference.

Independent checks.

- *Genus one*. For the 622 nonsingular cubics $m^2 = n^3 + an + b$ with $|a|, |b| \leq 12$, and for 400 further non-monic or shifted cubics and cubes $m^3 = an^2 + bn + c$, Sage’s integral points agree with our exact scans. Each family is reduced to an integral Weierstrass model with integrality tracked, and the Sage points are then pulled back. The certified lists contain hits far beyond any scan: $n^3 + n + 4$ at $n = 4128$, $n^3 + 2n^2 - 3n - 1$ at $n = 47882$, $6n^2 + n + 1 = 61301^3$ at $n = 6,196,204$, and $6n^2 - 7n - 6 = 95339^3$ at $n = 12,017,947$. A plain-Python gate re-verifies every stored point and pull-back.
- *Mordell census*. For all 20000 curves $y^2 = x^3 + k$, $0 < |k| \leq 10^4$, each row records the rank proof, generators, saturation index, point list and hash, and a scan to $|x| \leq 10^5$. A row whose scan disagrees cannot be labelled certified; the gate checks the domain and every row.
- *Theorem 4.4*. For $d, \deg F \leq 12$, Singular’s normalisation genus, and Sage’s count of places at infinity where it finishes, are compared with $\chi = d'(1 - S)$.
- *Adversarial test of the enumerator*. Rigid polynomials with hits planted up to $n \approx 10^9$ were constructed, and all planted hits were found in 220 trials.

Families that are neither certified in Lean nor independently cross-validated are labelled as scan evidence only, and are scanned exactly to 10^8 .

What remains.

- Effective enumeration in the finite type of genus at least 2 outside the Runge case, and quartic genus-one models.
- The exact Pell asymptotic $\kappa \log N + O(1)$ in Lean (the $O(\log N)$ bound is compiled), and the geometric half of Theorem 4.4.
- Analogues for non-polynomial sequences, and the Bilu–Tichy generalisation $F(x) = G(y)$.

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